

$$\begin{aligned} & \mathbf{g}_2^2 + \frac{1}{16\pi^2} \left(\frac{1}{12} \mathbf{g}_2^2 \left(3 \mathbf{g}_1^2 - 5 \mathbf{g}_2^2 \right) + \frac{2}{9} \mathbf{g}_2^4 \left(C_{\phi 1^2} + C_{\phi 2^2} \right) \mathbf{LF}_{3,0} [\mathbf{m}_2] + \frac{1}{3} \mathbf{g}_2^4 \left(C_{\phi 1^2} + C_{\phi 2^2} \right) \mathbf{LF}_{4,-1} [\mathbf{m}_2] - \right. \\ & \frac{15}{45} \mathbf{g}_2^4 \left(C_{\phi 1^2} + C_{\phi 2^2} \right) \mathbf{LF}_{5,-2} [\mathbf{m}_2] + \frac{1}{4} \left(-2 \mathbf{g}_2^2 \overline{\mathbf{y}}_e^{\text{PR}} \mathbf{y}_e^{\text{PR}} + \sum_{\mathbf{p}} \mathbf{g}_2^4 \right) \mathbf{LF}_{2,0} [\mathbf{m}_1^{\text{P}}] + \\ & \frac{1}{36} \mathbf{g}_2^2 \left(C_{\phi 1^2} + C_{\phi 2^2} \right) \left(-18 \overline{\mathbf{y}}_e^{\text{PR}} \mathbf{y}_e^{\text{PR}} + 13 \sum_{\mathbf{p}} \mathbf{g}_2^4 \right) \mathbf{LF}_{3,0} [\mathbf{m}_1^{\text{P}}] + \\ & \frac{1}{24} \mathbf{g}_2^2 \left(C_{\phi 1^2} + C_{\phi 2^2} \right) \left(12 \overline{\mathbf{y}}_e^{\text{PR}} \mathbf{y}_e^{\text{PR}} - 11 \sum_{\mathbf{p}} \mathbf{g}_2^4 \right) \mathbf{LF}_{4,-1} [\mathbf{m}_1^{\text{P}}] + \\ & \frac{4}{45} \sum_{\mathbf{p}} \mathbf{g}_2^4 \left(C_{\phi 1^2} + C_{\phi 2^2} \right) \mathbf{LF}_{5,-2} [\mathbf{m}_1^{\text{P}}] + \frac{3}{4} \mathbf{g}_2^2 \left(-2 \overline{\mathbf{y}}_d^{\text{PR}} \mathbf{y}_d^{\text{PR}} - 2 \overline{\mathbf{y}}_u^{\text{PR}} \mathbf{y}_u^{\text{PR}} + \sum_{\mathbf{p}} \mathbf{g}_2^4 \right) \mathbf{LF}_{2,0} [\mathbf{m}_q^{\text{P}}] + \\ & \frac{1}{12} \mathbf{g}_2^2 \left(C_{\phi 1^2} + C_{\phi 2^2} \right) \left(-18 \left(\overline{\mathbf{y}}_d^{\text{PR}} \mathbf{y}_d^{\text{PR}} + \overline{\mathbf{y}}_u^{\text{PR}} \mathbf{y}_u^{\text{PR}} \right) + 13 \sum_{\mathbf{p}} \mathbf{g}_2^4 \right) \mathbf{LF}_{3,0} [\mathbf{m}_q^{\text{P}}] + \\ & \frac{1}{8} \mathbf{g}_2^2 \left(C_{\phi 1^2} + C_{\phi 2^2} \right) \left(12 \overline{\mathbf{y}}_d^{\text{PR}} \mathbf{y}_d^{\text{PR}} + 12 \overline{\mathbf{y}}_u^{\text{PR}} \mathbf{y}_u^{\text{PR}} - 11 \sum_{\mathbf{p}} \mathbf{g}_2^4 \right) \mathbf{LF}_{4,-1} [\mathbf{m}_q^{\text{P}}] + \\ & \frac{4}{15} \sum_{\mathbf{p}} \mathbf{g}_2^4 \left(C_{\phi 1^2} + C_{\phi 2^2} \right) \mathbf{LF}_{5,-2} [\mathbf{m}_q^{\text{P}}] + \frac{1}{9} \mathbf{g}_2^4 \left(C_{\phi 1^2} + C_{\phi 2^2} \right) \mathbf{LF}_{3,0} [\tilde{\mu}] + \\ & \frac{1}{6} \mathbf{g}_2^4 \left(C_{\phi 1^2} + C_{\phi 2^2} \right) \mathbf{LF}_{4,-1} [\tilde{\mu}] - \frac{8}{45} \mathbf{g}_2^4 \left(C_{\phi 1^2} + C_{\phi 2^2} \right) \mathbf{LF}_{5,-2} [\tilde{\mu}] + \mathbf{g}_1^4 \tilde{\mu}^2 \mathbf{LF}_{2,-2,-1} [\mathbf{m}_1, \tilde{\mu}] + \\ & \frac{1}{2} \mathbf{m}_1 \tilde{\mu} \mathbf{g}_1^4 \left(\overline{C_{\phi 1 \phi 2}} + C_{\phi 1 \phi 2} \right) \mathbf{LF}_{2,2,0} [\mathbf{m}_1, \tilde{\mu}] - \frac{4}{3} \mathbf{g}_2^4 \left(C_{\phi 1^2} + C_{\phi 2^2} \right) \mathbf{LF}_{2,1,0} [\mathbf{m}_2, \tilde{\mu}] - \\ & 2 \mathbf{g}_2^4 \mathbf{LF}_{2,2,-2} [\mathbf{m}_2, \tilde{\mu}] + \frac{1}{3} \mathbf{g}_2^4 \left(2 \left(C_{\phi 1^2} + C_{\phi 2^2} + 3 \mathbf{m}_2^2 \right) + 9 \tilde{\mu}^2 \right) \mathbf{LF}_{2,2,-1} [\mathbf{m}_2, \tilde{\mu}] + \\ & \frac{1}{6} \mathbf{m}_2 \mathbf{g}_2^4 \left(-6 \mathbf{m}_2 \left(C_{\phi 1^2} + C_{\phi 2^2} \right) + 13 \tilde{\mu} \left(\overline{C_{\phi 1 \phi 2}} + C_{\phi 1 \phi 2} \right) \right) \mathbf{LF}_{2,2,0} [\mathbf{m}_2, \tilde{\mu}] + \\ & \frac{2}{3} \mathbf{g}_2^4 \left(C_{\phi 1^2} + C_{\phi 2^2} \right) \mathbf{LF}_{3,1,-1} [\mathbf{m}_2, \tilde{\mu}] - \frac{10}{3} \mathbf{g}_2^4 \left(C_{\phi 1^2} + C_{\phi 2^2} \right) \mathbf{LF}_{3,2,-2} [\mathbf{m}_2, \tilde{\mu}] + \\ & \frac{1}{3} \mathbf{m}_2 \mathbf{g}_2^4 \left(9 \mathbf{m}_2 \left(C_{\phi 1^2} + C_{\phi 2^2} \right) - \tilde{\mu} \left(\overline{C_{\phi 1 \phi 2}} + C_{\phi 1 \phi 2} \right) \right) \mathbf{LF}_{3,2,-1} [\mathbf{m}_2, \tilde{\mu}] + \\ & 2 \mathbf{g}_2^4 \left(C_{\phi 1^2} + C_{\phi 2^2} \right) \mathbf{LF}_{3,3,-3} [\mathbf{m}_2, \tilde{\mu}] - 2 \mathbf{g}_2^4 \mathbf{m}_2^2 \left(C_{\phi 1^2} + C_{\phi 2^2} \right) \mathbf{LF}_{3,3,-2} [\mathbf{m}_2, \tilde{\mu}] + \\ & 2 \mathbf{g}_2^4 \left(C_{\phi 1^2} + C_{\phi 2^2} \right) \mathbf{LF}_{4,2,-3} [\mathbf{m}_2, \tilde{\mu}] - 2 \mathbf{g}_2^4 \mathbf{m}_2^2 \left(C_{\phi 1^2} + C_{\phi 2^2} \right) \mathbf{LF}_{4,2,-2} [\mathbf{m}_2, \tilde{\mu}] + \\ & 3 \overline{\mathbf{y}}_d^{\text{PR}} \mathbf{y}_d^{\text{SR}} \overline{\mathbf{y}}_u^{\text{ST}} \mathbf{y}_u^{\text{PT}} \mathbf{LF}_{1,1,0} [\mathbf{m}_d^{\text{r}}, \mathbf{m}_0^{\text{t}}] - \mathbf{g}_2^2 \overline{\mathbf{a}}_e^{\text{PR}} \mathbf{a}_e^{\text{PR}} \mathbf{LF}_{2,1,0} [\mathbf{m}_1^{\text{P}}, \mathbf{m}_e^{\text{r}}] + \\ & \frac{1}{2} \mathbf{g}_2^2 \left(\overline{\mathbf{a}}_e^{\text{PR}} \mathbf{a}_e^{\text{PR}} + \tilde{\mu}^2 \overline{\mathbf{y}}_e^{\text{PR}} \mathbf{y}_e^{\text{PR}} \right) \mathbf{LF}_{3,1,-1} [\mathbf{m}_1^{\text{P}}, \mathbf{m}_e^{\text{r}}] - \\ & \mathbf{g}_2^2 \left(\overline{C_{\phi 1 \phi 2}} \tilde{\mu} \overline{\mathbf{y}}_e^{\text{PR}} \mathbf{a}_e^{\text{PR}} + \overline{\mathbf{a}}_e^{\text{PR}} \left(2 C_{\phi 2^2} \mathbf{a}_e^{\text{PR}} + C_{\phi 1 \phi 2} \tilde{\mu} \mathbf{y}_e^{\text{PR}} \right) \right) \mathbf{LF}_{3,1,0} [\mathbf{m}_1^{\text{P}}, \mathbf{m}_e^{\text{r}}] + \\ & \frac{1}{4} \mathbf{g}_2^2 \left(\overline{\mathbf{a}}_e^{\text{PR}} \left(18 C_{\phi 2^2} \mathbf{a}_e^{\text{PR}} + 13 C_{\phi 1 \phi 2} \tilde{\mu} \mathbf{y}_e^{\text{PR}} \right) + \tilde{\mu} \overline{\mathbf{y}}_e^{\text{PR}} \left(13 \overline{C_{\phi 1 \phi 2}} \mathbf{a}_e^{\text{PR}} + 2 \tilde{\mu} \mathbf{y}_e^{\text{PR}} \left(4 C_{\phi 1^2} + C_{\phi 2^2} \right) \right) \right) \\ & \mathbf{LF}_{4,1,-1} [\mathbf{m}_1^{\text{P}}, \mathbf{m}_e^{\text{r}}] - \frac{1}{3} \mathbf{g}_2^2 \left(\overline{\mathbf{a}}_e^{\text{PR}} \left(\mathbf{a}_e^{\text{PR}} \left(C_{\phi 1^2} + 7 C_{\phi 2^2} \right) + 7 C_{\phi 1 \phi 2} \tilde{\mu} \mathbf{y}_e^{\text{PR}} \right) + \right. \\ & \left. \tilde{\mu} \overline{\mathbf{y}}_e^{\text{PR}} \left(7 \overline{C_{\phi 1 \phi 2}} \mathbf{a}_e^{\text{PR}} + \tilde{\mu} \mathbf{y}_e^{\text{PR}} \left(7 C_{\phi 1^2} + C_{\phi 2^2} \right) \right) \right) \mathbf{LF}_{5,1,-2} [\mathbf{m}_1^{\text{P}}, \mathbf{m}_e^{\text{r}}] - \\ & 3 \mathbf{g}_2^2 \overline{\mathbf{a}}_d^{\text{PR}} \mathbf{a}_d^{\text{PR}} \mathbf{LF}_{2,1,0} [\mathbf{m}_q^{\text{P}}, \mathbf{m}_d^{\text{r}}] + \frac{3}{2} \mathbf{g}_2^2 \left(\overline{\mathbf{a}}_d^{\text{PR}} \mathbf{a}_d^{\text{PR}} + \tilde{\mu}^2 \overline{\mathbf{y}}_d^{\text{PR}} \mathbf{y}_d^{\text{PR}} \right) \mathbf{LF}_{3,1,-1} [\mathbf{m}_q^{\text{P}}, \mathbf{m}_d^{\text{r}}] - \\ & 3 \mathbf{g}_2^2 \left(\overline{C_{\phi 1 \phi 2}} \tilde{\mu} \overline{\mathbf{y}}_d^{\text{PR}} \mathbf{a}_d^{\text{PR}} + \overline{\mathbf{a}}_d^{\text{PR}} \left(2 C_{\phi 2^2} \mathbf{a}_d^{\text{PR}} + C_{\phi 1 \phi 2} \tilde{\mu} \mathbf{y}_d^{\text{PR}} \right) \right) \mathbf{LF}_{3,1,0} [\mathbf{m}_q^{\text{P}}, \mathbf{m}_d^{\text{r}}] + \\ & \frac{3}{4} \mathbf{g}_2^2 \left(\overline{\mathbf{a}}_d^{\text{PR}} \left(18 C_{\phi 2^2} \mathbf{a}_d^{\text{PR}} + 13 C_{\phi 1 \phi 2} \tilde{\mu} \mathbf{y}_d^{\text{PR}} \right) + \tilde{\mu} \overline{\mathbf{y}}_d^{\text{PR}} \left(13 \overline{C_{\phi 1 \phi 2}} \mathbf{a}_d^{\text{PR}} + 2 \$$